Pre-Threshold Graduated Deleveraging for Perpetual Futures: Proposal and Evaluation Framework

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Abstract

Hyperliquid already stages large-position liquidation at the maintenance threshold (20% per round, 30-second cooldown). We propose graduated deleveraging in a warning zone *above* that threshold, shrinking positions before the existing engine fires. Paired Monte Carlo across four regimes shows the mechanism cuts threshold-entry notional by roughly half in slow cascades but is inert in fast gaps. Required vault capital falls 26–44% under regime-conditional sizing, yet this benefit vanishes under worst-case cross-regime sizing because the binding constraint is the gap regime. The paper defines the mechanism, reports simulation results, and states the conditions under which the backstop-funding argument holds.

Introduction

On Hyperliquid, liquidation proceeds through four stages when equity falls below maintenance margin [1]: (1) market liquidation (full for ≤\$100K, 20%/round for larger); (2) 30-second cooldown then full remainder; (3) backstop vault at $\frac{2}{3}$ maintenance margin; (4) ADL if the vault cannot absorb [2, 3]. The 20%/cooldown mechanism already mitigates cliff risk for large individual positions.

Two sources of cliff risk persist. **Cross-account clustering:** many accounts hit maintenance margin at the same price; aggregate flow fires simultaneously even though each position is individually staged. **Sub-\$100K positions:** liquidated in full at threshold, generating large aggregate notional during broad selloffs.

We propose graduated deleveraging: partial position closure in a warning zone above the maintenance threshold, upstream of the existing engine. The protocol cares because this may reduce aggregate threshold-entry flow. Deployers care because reduced flow may lower vault demand and idle capital—but only if the event-type distribution is favorable. Section 1 presents the protocol proposal, Section 2 analyzes the deployer benefit, Section 3 reports simulation evidence and deployment constraints.

1. Proposal for the Protocol

The objective is to reduce cross-account threshold-entry flow before the existing staged-liquidation engine activates.

1.1. Definition

Convention: the distressed position is **long**; shorts reverse the inequalities.

Definition 1 (Liquidation and warning prices)**.** Position i: notional S_i, leverage ℓ_i, maintenance margin rate m, entry price p_0.

$$p_i^{\text{liq}} = p_0\big(1-\tfrac{1}{\ell_i}+m\big), \qquad p_i^{\text{warn}} = p_0\big(1-\tfrac{1}{\ell_i}+wm\big), \quad w > 1.$$

Warning zone: $\big[p_i^{\text{liq}},\, p_i^{\text{warn}}\big]$, width $(w-1)\, m\, p_0$.

Definition 2 (Normalized depth)**.**

$$\phi_i(p) \;=\; \frac{p_i^{\text{warn}} - p}{p_i^{\text{warn}} - p_i^{\text{liq}}} \;\in\; [0,1].$$

$\phi = 0$ at warning boundary; $\phi = 1$ at liquidation boundary.

Each block, for each position in its warning zone, the protocol submits a close order for fraction $\alpha \cdot \phi_i(p)$ of the remaining position, subject to the control policy (§3.7).

Price source. Same mark price as the liquidation engine. No new oracle. Fires earlier, so more sensitive to transient dislocations.

1.2. Episode Taxonomy

Definition 3 (Warning-zone episode)**.** A maximal contiguous interval of blocks during which mark price remains in position i's warning zone.

Definition 4 (Episode types)**.** *Recovery*: mark exits above p_i^{warn}. *Liquidation*: mark exits below p_i^{liq}. *Gap*: mark jumps from above p_i^{warn} to below p_i^{liq} in one block. Exhaustive and mutually exclusive.

Recovery episodes generate false-positive costs. Liquidation episodes generate the benefit. Gap events render the mechanism inert.

1.3. Execution Framework

Let $t = 0, 1, \ldots$ index blocks within a warning-zone episode.

$R_t \in (0,1]$	remaining fraction at block t
$\phi_t \in [0,1]$	warning-zone depth at block t
$a_t \in \{0,1\}$	activation indicator (§3.7)
$f_t \in [0,1]$	realized fill fraction at block t
T	stopping time (episode end)

Position dynamics:

$$R_{t+1} \;=\; R_t\,\big(1 - a_t\,\alpha\,\phi_t\,f_t\big), \qquad R_0 = 1. \tag{1}$$

Surviving fraction at episode end:

$$\bar{\beta} \;=\; R_T \;=\; \prod_{t=0}^{T-1}\big(1 - a_t\,\alpha\,\phi_t\,f_t\big). \tag{2}$$

$\{p_t\}$ and $\{f_t\}$ are **exogenous inputs**: the recursion takes them as given and produces $\{R_t\}$ and $\bar{\beta}$. Both are stochastic, venue-specific, and require historical data to estimate.

1.4. First-Order Properties

Let Q = aggregate hard-liquidation notional, D = order-book depth (exogenous), C = vault capital.

$$\mathrm{ADL}(\Delta p, C) = \big(Q - D - C\big)^+, \tag{3}$$

$$\mathrm{ADL}^G(\Delta p, C) = \big(\bar\beta\, Q - D - C\big)^+. \tag{4}$$

These compress Hyperliquid's multi-stage waterfall into aggregate absorption capacity $D+C$; they are not a block-by-block simulation.

Observation 1 (Monotonicity (identity))**.** $\bar\beta \le 1 \Rightarrow \mathrm{ADL}^G \le \mathrm{ADL}$.

Whether $\bar\beta < 1$ obtains, and at what cost, is separate.

Interaction with staged liquidation. Pre-threshold reduction changes not only the size but also *which branch* of the staged engine applies. A \$200K position reduced to \$80K before threshold falls below the \$100K staged threshold, so it is liquidated in full, but at \$80K instead of \$200K. The interaction is not purely multiplicative.

Exogenous-D limitation. Graduated orders consume depth above the liquidation level. If this depth does not replenish, effective D at threshold is reduced. The net effect is regime-dependent (§2.2).

2. Benefit for the Deployer

Does the mechanism reduce residual vault demand and required idle capital? The benefit exists in slow-cascade regimes and disappears under worst-case sizing when fast-gap risk binds.

2.1. Welfare

$$\Delta W = B_{\mathrm{ADL}} - C_{\mathrm{FP}} - C^d_{\mathrm{EX}} - C^i_{\mathrm{EX}}. \tag{5}$$

$B_{\mathrm{ADL}} \ge 0$**: ADL harm avoided.** $B_{\mathrm{ADL}} = \Delta\mathrm{ADL} \cdot h_{\mathrm{ADL}}$. Borne by traders spared ADL.

$C_{\mathrm{FP}} \ge 0$**: False-positive cost.** $C_{\mathrm{FP}} = \sum_{\mathrm{recovery}}(1 - R_T) \cdot S_i \cdot h_{\mathrm{FP}}$. Borne by near-distressed traders whose positions recover.

$C^d_{\mathrm{EX}} \ge 0$**: Direct execution cost.** $C^d_{\mathrm{EX}} = \sum Q^{(j)}_G \cdot \Delta s^{(j)}_d$. Borne by graduated traders.

$C^i_{\mathrm{EX}} \ge 0$**: Indirect market-quality cost.** Spread widening and depth withdrawal from anticipated flow. Borne by all participants.

$\Delta W > 0$ is not a Pareto improvement: beneficiaries differ from cost-bearers.

2.2. Regime Dependence

Deployer relevance depends on the prevailing market regime.

2.2.1. Regime A: Slow Cascade, Replenishing Depth

Price declines over many blocks; depth replenishes.
$\bar\beta$: low. C_{FP}: low. $C^d_{\mathrm{EX}}, C^i_{\mathrm{EX}}$: low.
Most plausibly positive welfare.
Applies to liquid markets with moderate-speed selloffs and clustered thresholds.

2.2.2. Regime B: Fast Gap, No Fill

Price gaps past the warning zone in 1–3 blocks.
$\bar\beta \approx 1$. All terms ≈ 0.
Neutral by construction.
Applies to earnings gaps, geopolitical shocks, weekend gaps. Leverage caps and dynamic margin are the appropriate controls.

2.2.3. Regime C: Choppy, Mean-Reverting

Price oscillates around the warning boundary; recovery episodes dominate.
C_{FP}: high. **Most plausibly negative.**
Applies to range-bound markets with high volatility relative to zone width.

2.2.4. Regime D: Thin Book, Protocol Flow Dominates

Graduated flow is large relative to depth.
$C^d_{\mathrm{EX}}, C^i_{\mathrm{EX}}$: high. **Ambiguous.**
Applies to illiquid assets, early deployments, off-hours. Throttling or auctions are preferable.

2.3. Backstop Funding

Capital efficiency and welfare are distinct.

HIP-3 deployers fund their own backstop vaults. Capital in the vault is idle. At backstop trigger (equity $< \frac{2}{3}$ maintenance margin), the vault absorbs the position and the user's remaining equity, but must post full maintenance margin to hold it, creating a per-position capital gap of $N/(6\ell)$. The required vault balance depends on the distribution of aggregate capital gaps across stress events. Implementation requires both ex ante sizing and live monitoring of vault state, liquidation proximity, and market depth.

2.3.1. Risk-Target Capital Sizing

Let R denote the random residual notional reaching the vault.

Definition 5 (Required capital)**.** For tolerated ADL probability $\eta \in (0,1)$:

$$C^*(\eta) = \inf\big\{C \ge 0 : \Pr(R > C) \le \eta\big\} = F_R^{-1}(1-\eta). \tag{6}$$

Definition 6 (Capital intensity)**.**

$$\gamma(\eta) = C^*(\eta)\,/\,N, \tag{7}$$

where N is total notional at risk.

If the graduated layer shifts F_R leftward, $C^*(\eta)$ falls for every η.

2.3.2. Worst-Case Cross-Regime Sizing

A deployer sizing against the worst regime faces:

$$C^*_{\mathrm{wc}}(\eta) = \max_k\ F^{-1}_{R_k}(1-\eta), \tag{8}$$

where R_k is vault demand in regime k. When Regime B binds, the mechanism is inert (F_{R_B} identical in both modes):

$$C^*_{\mathrm{wc,graduated}}(\eta) = C^*_{\mathrm{wc,baseline}}(\eta). \tag{9}$$

The capital benefit requires: (i) slow-cascade events dominate the asset's history; (ii) the deployer sizes vault to the regime-conditional distribution; (iii) vault demand is large enough that some events produce overflow. Otherwise, the mechanism provides no capital relief.

2.3.3. Scaling Heuristic (Supplementary)

Under simplified assumptions (linear impact, diffusion dynamics, depth proportional to OI):

$$\gamma \approx \frac{\beta_{\text{frac}}^2}{2k\epsilon} + \left(\sigma\sqrt{\tau} \cdot z_{1-\alpha} - \tfrac{2}{3}m\right)\beta_{\text{frac}}, \tag{10}$$

where β_{frac} is the OI fraction reaching the vault, k is calm-state depth ratio, ϵ is stress haircut, $\sigma\sqrt{\tau}$ is unwind volatility. Slippage scales as β_{frac}^2: halving β_{frac} cuts it by 75%.

This is a directional heuristic, not a production rule. Deeper books, leverage caps, and dynamic margin achieve comparable β_{frac} reductions. Use the quantile rule (6) for sizing.

2.3.4. Dynamic Fee Routing

A dynamic reinvestment rule parameterized by coverage ratio $\rho = C/C^*$:

$$\alpha_r(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot g(1-\rho), \tag{11}$$

with $g\colon [0,1] \to [0,1]$ non-decreasing, $g(0) = 0$, $g(1) = 1$. Fully funded ($\rho \geq 1$): $\alpha_r = \alpha_{\min}$. Depleted ($\rho \to 0$): $\alpha_r = \alpha_{\max}$.

If C^* falls under regime-conditional sizing, the vault reaches target faster. This benefit shares the same conditionality: it does not hold under worst-case sizing.

3. Evaluation and Deployment

3.1. Monte Carlo Design

For each seed, the exogenous market path and account population are held fixed; only the graduated layer changes. Both runs use the exact Hyperliquid baseline: \$100K staged threshold, 20% partial rounds, 30-second cooldown, vault takeover at $\frac{2}{3}$ maintenance margin, ADL.

Phase 1 (200 realizations × 4 regimes, 4 accounts): tests whether the mechanism reduces threshold-entry flow.

Phase 2 (100 realizations × 6 profiles × 4 regimes = 2,400 runs): tests the deployer-capital story. Sweeps population size (10, 50, 100 accounts), threshold clustering (±0.3% to ±2.5% entry spreads), and vault capital (2–100% of notional). Uses a hybrid price path: slow cascade through the warning zone, then a −4% gap below p^{liq}. The cascade gives the graduated engine time to act; the gap triggers vault takeover on remaining positions, making vault demand proportional to R at the gap.

Post-hoc vault sweep. Vault capital does not affect liquidation dynamics; it only determines the vault/ADL split. Vault demand $V = V_{\text{absorbed}} + V_{\text{ADL}}$ is independent of capital level. For any C: $\text{ADL}(C) = (V - C)^+$. One run per seed suffices for the full capital sweep.

Three-layer evaluation. Each realization receives three independent verdicts:

Threshold layer	Does Δ threshold-entry exceed 0.5% of total notional?
Downstream layer	Does vault demand change by >0.5%?
Capital layer	Is vault demand nonzero in either run?

At the regime level, the capital channel is labeled *active* (>10% shift in required capital), *marginal* (≤10%), or *inactive* (vault demand ≈0). These are reporting conventions for the simulation, not general categories.

All results are model-based and conditional on synthetic inputs. They are not evidence about real Hyperliquid behavior.

3.2. Phase 1 Results

	helps	inert	harms	Δ thresh.	$\Delta\bar{\beta}$
A	100%	0%	0%	−\$301K	−0.52
B	0%	100%	0%	\$0	0.00
C	84.5%	0%	15.5%	−\$315K	−0.35
D	100%	0%	0%	−\$151K	−0.26

A: $\bar{\beta}$ drops from 1.0 to 0.47; 53% pre-closed. False-positive cost negligible (\$39). Structural result; magnitude assumption-sensitive.

B: All deltas zero. Gap bypasses the zone; $\bar{\beta} = 1$ in both runs.

C: Threshold reduction (−\$315K) partly offset by false-positive cost (\$263K). Cost exceeds benefit in 15.5% of runs.

D: Threshold benefit persists (−\$151K) but smaller; poor fills limit pre-closing ($\bar{\beta}$: 1.0 → 0.74).

Phase 1 gap. $\Delta\text{vault} = \Delta\text{ADL} = 0$ in all regimes. Vault triggers at equity $< \frac{2}{3}$ maintenance margin, roughly 1% below p^{liq}. In gradual cascades, the threshold engine closes most positions before reaching that level. This motivates the hybrid path in Phase 2.

3.3. Phase 2 Results

With 100 accounts and high clustering (representative):

	Threshold	Downstream	Capital
A	helps	helps	**active**
B	inert	neutral	marginal
C	helps (FP)	neutral	inactive
D	helps	helps	marginal

A: Vault demand drops from \$13.2M to \$6.0M (mean, −54%). $C^*_{0.05}$ falls from \$14.8M to \$10.9M (−26%). With low clustering: −44%.

B: Zero effect at all layers.

C: Threshold benefit exists but vault never triggers; capital channel inactive.

D: (22% fills, \$30K depth) Threshold benefit persists; capital shift only −4% to −5%.

Figure 1 summarizes the two-layer distinction: Panel A shows where the mechanism helps or costs at the threshold layer; Panel B shows where that translates (or fails to translate) into deployer-capital benefit.

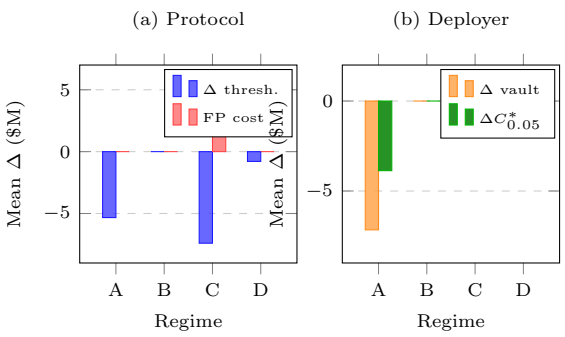

Figure 1: **(a)** Threshold-entry reduction and false-positive cost. **(b)** Vault-demand and required-capital shift. Only Regime A shows an active capital channel. 100 accounts, high clustering.

3.4. Cross-Regime Capital Constraint

Deployers cannot condition vault capital on regime. Required capital is the worst case across regimes:

Regime	$C^*_{0.05}$ baseline	$C^*_{0.05}$ graduated
A	\$14.8M	\$10.9M
B	**\$14.8M**	**\$14.8M**
C	\$0	\$0
D	\$14.6M	\$14.0M
max	\$14.8M (B)	\$14.8M (B)

Regime B binds. Worst-case required capital is unchanged: $\Delta C^*_{\text{wc}} = 0$. This holds across all population sizes and clustering levels tested (Figure 3).

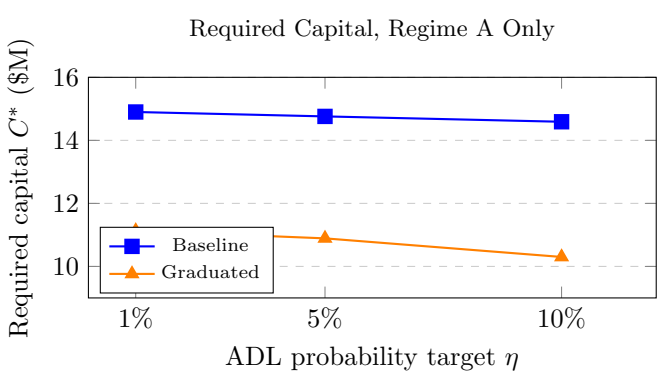

Figure 2: $C^*(\eta)$ for $\Pr(\text{ADL}) \leq \eta$, Regime A (100 accounts, high clustering). Graduated layer shifts the curve down 25–29%. Conditional on Regime A; does not survive cross-regime sizing (§3.4).

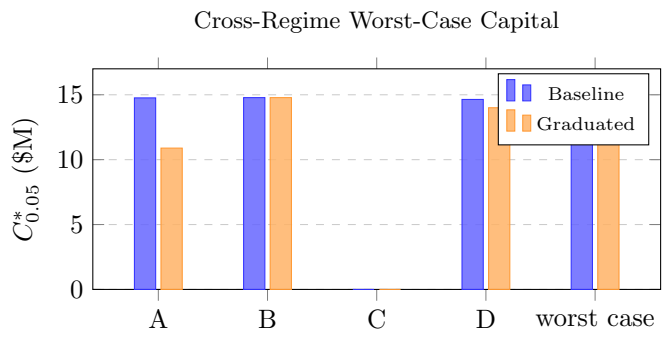

Figure 3: $C^*_{0.05}$ by regime and worst case. Regime A shifts down. B does not move. Worst case pinned by B: cross-regime capital unchanged. 100 accounts, high clustering.

3.5. Event-Mix Sensitivity

Whether the capital benefit is accessible depends on the deployer's event-type mix. Consider a deployer whose events are fraction p slow-cascade (Regime A) and $1-p$ fast-gap (Regime B). At ADL target η, the required capital under the mixture is:

$$C^*_{\text{mix}}(\eta, p) = F^{-1}_{p\,R_A + (1-p)\,R_B}(1-\eta).$$

Since each Regime B event produces vault demand ≈\$14.8M (inert to the mechanism), the graduated layer can only reduce C^*_{mix} once $(1-p) < \eta$, i.e. once gap events are rarer than the ADL target. At $\eta = 5\%$, the capital benefit appears only when $p > 95\%$. Figure 4 plots this relationship.

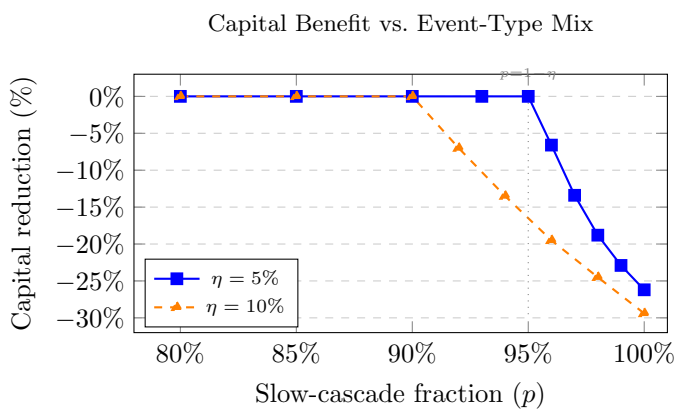

Figure 4: Capital reduction vs. event mix. Benefit is zero until slow-cascade fraction exceeds $1-\eta$. At η=5%: benefit requires >95% slow cascades.

3.6. Alternatives

The baseline is Hyperliquid's existing waterfall. The proposal adds a pre-threshold layer upstream.

Dynamic margin and leverage caps address root cause; prefer if commercially viable.

Staged liquidation (20%/cooldown) manages per-account execution. Graduated deleveraging manages cross-account aggregate flow. They compose.

Liquidation auctions address execution quality; preferable in deep markets.

Dynamic throttling manages per-block flow; more robust to manipulation; composes with graduated deleveraging.

Graduated deleveraging is a second-best upstream flow-smoothing mechanism for venues that cannot control leverage directly.

3.7. Operational Design

3.7.1. Control Policy (Draft)

Activation. $a_t = 1$ after $n_h \geq 1$ consecutive blocks in zone.
Pause. $a_t = 0$ if mark/index divergence $> \theta$.
Close sizing. $\alpha_{\text{eff}} \cdot \phi_t \cdot R_t \cdot S_i$, where $\alpha_{\text{eff}} = \alpha \cdot \min(1, D_{\text{near}}/D_{\text{min}})$.
Order type. Marketable limit, slippage tolerance δ_s.
Fill handling. Partial fills close filled portion; no-fills leave R_t unchanged.
Scheduling. Up to K positions per block, ϕ_t descending.
Deactivation. Reset after m blocks above p_i^{warn}.

3.7.2. Failure Modes

1. Quote fading. MMs pull bids ahead of predictable flow. Mitigation: adaptive α_{eff}.
2. Adverse selection. Order direction/timing inferable. Mitigation: slippage cap.
3. Griefing. Attacker pushes mark into zone. Mitigation: divergence pause, hysteresis.
4. Nuisance deleveraging. Erosion across recoveries. Mitigation: hysteresis, minimum size.
5. Depth depletion. Graduated flow reduces depth at threshold. Mitigation: adaptive α_{eff}.
6. Throughput saturation. Cap K insufficient in broad cascades. Mitigation: higher K, priority scheduling.
7. Gap ineffectiveness. Most damaging events bypass the zone. No mitigation within this mechanism.

3.8. Measurement and Deployment

Required before deployment:

$\bar{\beta}$**:** dwell-time distributions; episode-type fractions; fill rates near distress; depth replenishment speed. B_{ADL}**:** ADL frequency/volume per asset; per-dollar harm. C_{FP}**:** recovery-episode frequency; per-dollar false-positive harm. $C^d_{\text{EX}}, C^i_{\text{EX}}$**:** spread/slippage near liquidations; depth changes during stress. **Backstop funding:** vault drawdown distribution; residual-risk distribution; fee revenue as fraction of OI. **Regime classification:** historical gap-event fraction.

Simulation baseline. Must implement the real Hyperliquid waterfall ($100K threshold, 20%/cooldown, vault at $\frac{2}{3}$ MM, ADL). Comparison against a strawman overstates benefit.

Implementation unknowns. Exact vault acceptance logic, PnL treatment (whether profitable absorptions compound the buffer), simultaneous absorption sequencing, and margin-mode transfer mechanics under portfolio margin remain unresolved.

Sequence: measure → calibrate → simulate → evaluate ΔW and $\Delta C^*(\eta)$ → deploy.

Conclusion

Protocol takeaway. Graduated deleveraging reduces threshold-entry notional by roughly half in slow-cascade regimes, is inert in fast gaps, and incurs false-positive costs in choppy markets. It changes the liquidation path in a regime-dependent way.

Deployer takeaway. Regime-conditional required capital falls 26–44%, but this does not survive worst-case cross-regime sizing. If the binding regime is a fast gap, required capital is unchanged. The benefit is accessible to deployers whose assets are dominated by slow-cascade events, not to those who must size for gap risk.

Strong claim. Graduated deleveraging is plausibly attractive only where distress develops over multiple blocks, depth replenishes, and the event distribution permits regime-conditional sizing.

Strong non-claim. Nothing here constitutes evidence for implementation. The incremental benefit over Hyperliquid's existing staged liquidation requires venue-specific measurement, simulation, and controlled deployment.

References

[1] Hyperliquid. Liquidations. *Hyperliquid Documentation*, 2025.

[2] Autodeleveraging: Impossibilities and Optimization. *arXiv:2512.01112v2*, 2025.

[3] Autodeleveraging as Online Learning. *arXiv:2602.15182*, 2026.